TRANSMIGRATION EXECUTIVE DIALOGUE · PANEL II · JAKARTA, 24 AUGUST 2026

Where Indonesia's Next Growth Will Come From

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▸ BIO

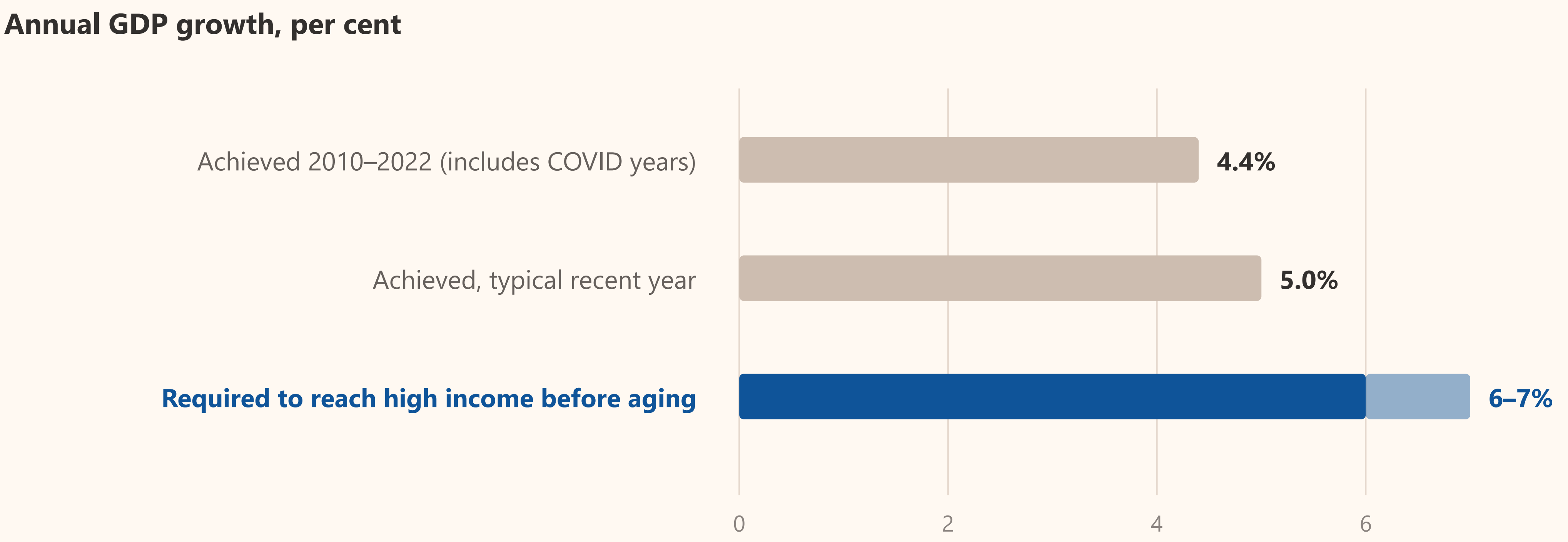
Evidence from three general-equilibrium studies: the five national growth strategies prepared for Bappenas, and two provinces at opposite ends of Indonesia's structural transformation. What they say about industrialisation, downstreaming, and building a productive economic area from a standing start.

THE PROBLEM

Indonesia is growing too slowly, and its population is aging

Indonesia needs 6–7% annual growth to reach high income before the demographic dividend closes. It has been growing at about 5%, and at 4.4% a year over 2010–2022.

> **Indonesia must lift growth by roughly two percentage points within the next decade, while its labour force is still growing.**



WHAT THIS REQUIRES

Faster growth has only two sources, and one traditional route

> **Over a ten-year horizon, growth comes from more production capacity and higher productivity. Nothing else.**

1

More production capacity

Capacity grows through investment and capital accumulation. Short-run demand management — consumption, government spending, exports — does not raise the medium-term growth path. **This is a supply-side problem.**

2

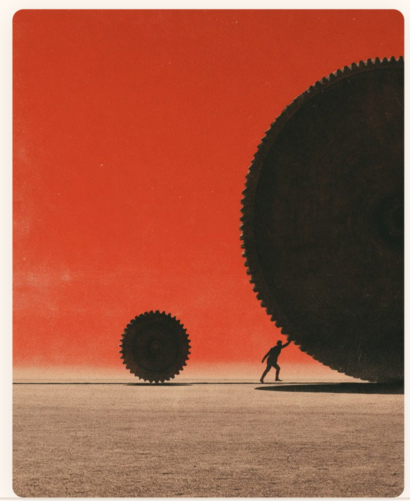
Higher productivity

Productivity rises through technology adoption, skills, and the spillovers that infrastructure and industry generate. In the models, productivity is consistently **the single largest source of additional growth**.

3

Structural transformation, but not the old route

Labour moving to more productive activities is still the mechanism. But the export-manufacturing route that worked for Korea, Taiwan and 1980s Indonesia is harder now: competition from China, automation, and re-shoring have produced **premature deindustrialisation** across the developing world.



THE NATIONAL EVIDENCE

Five growth strategies, tested in one model

Bappenas commissioned five white papers on how Indonesia might reach 8% growth. All five were translated into scenarios and simulated in the same CGE model.

> **Each strategy was run at a conservative and a progressive setting, separately and then all together.**

1

Infrastructure

Infrastructure investment growing **15% a year** (conservative) or **25%** (progressive), against about 10% over the last decade. Utilities, transport and information services.

2

Digitalisation and Industry 4.0

Technology adoption raising labour productivity by **0.9–3.4% a year** depending on the sector, from metals at the low end to machinery at the high end.

3

Natural-resource downstreaming

Upstream mineral exports diverted to domestic supply (**−8 to −12% a year**), downstream investment up **25% a year**, with technology spillover.

4

Agricultural transformation

Mechanisation and technology raising productivity of leading regional crops by **1–3% a year**, with processing capacity up **10–20% a year**.

5

Green and blue economy

Cheaper freight, energy efficiency, public transport and maritime investment. Judged on **emissions reduction without sacrificing growth**, not on growth.

6

The model

IndoTERM: bottom-up inter-regional CGE, 34 provinces, 185 commodities, dynamic capital accumulation, endogenous investment, CO_2 accounting. Built by CEDS Universitas Padjadjaran with Victoria University, ADB and Bappenas.

THE NATIONAL EVIDENCE

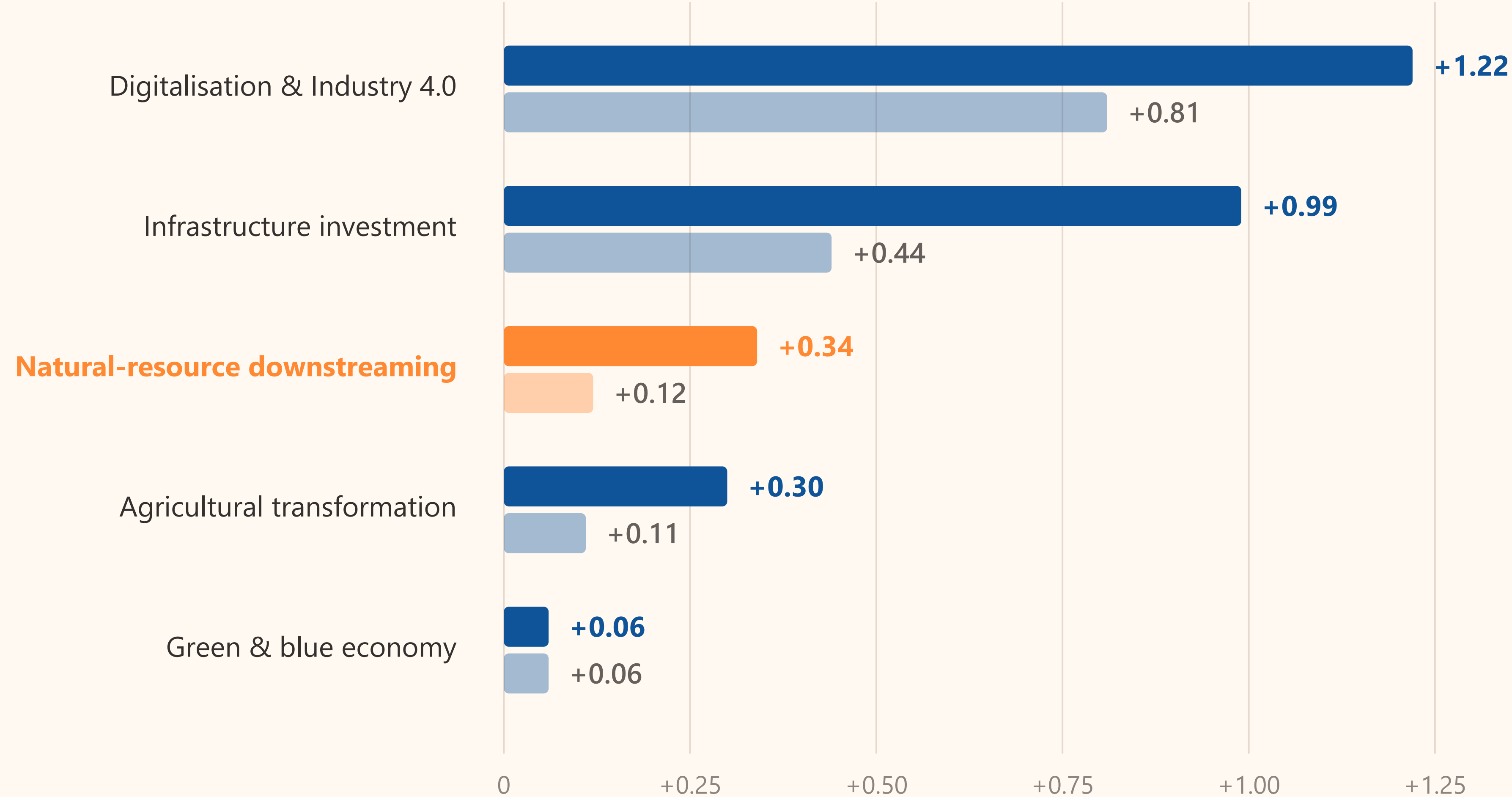
What each strategy adds to growth

Percentage points added to the annual growth rate, averaged over 2023–2034, each strategy simulated on its own.

> **Downstreaming raises growth, but on its own it is one of the smallest levers. Productivity — digitalisation and infrastructure — does most of the work.**

Percentage points added to annual growth, 2023–2034

baseline without any strategy: 5.10% a year · **conservative** and **progressive** scenarios

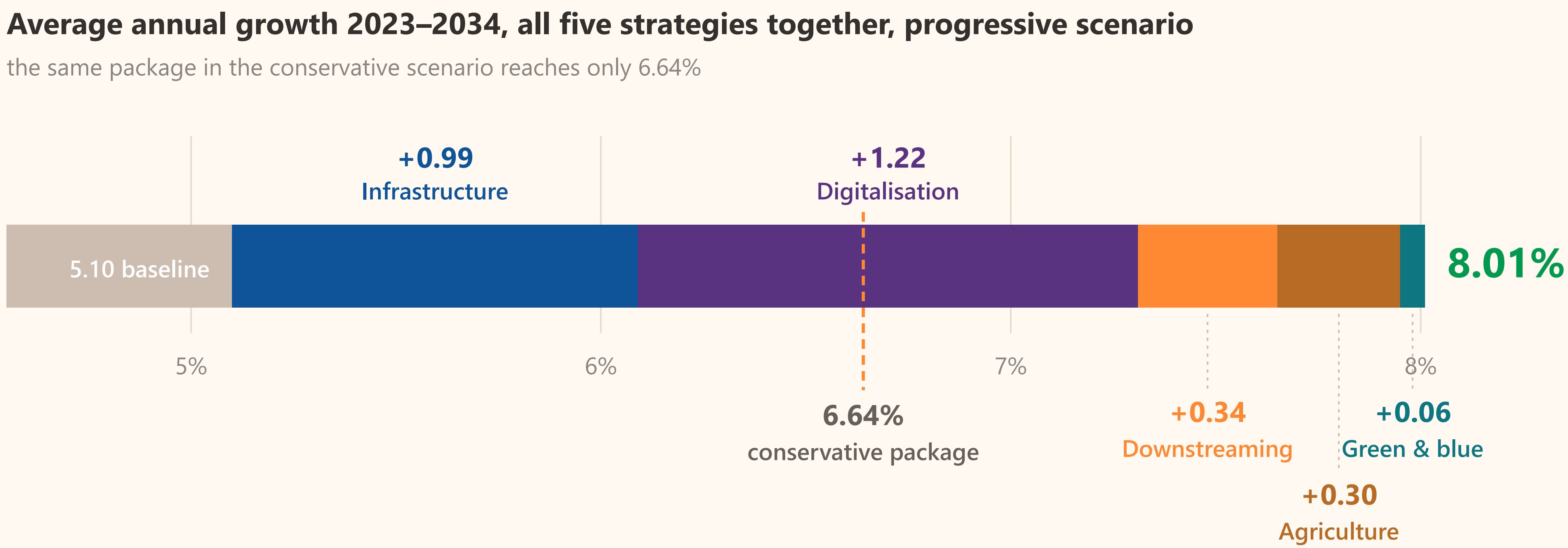


THE NATIONAL EVIDENCE

The route to 8 per cent

The five strategies combined, each at its most ambitious setting.

> **8% is reachable — but only with every strategy running at maximum at the same time. The conservative version of the same package reaches 6.6%.**



THE NATIONAL EVIDENCE

What the 8 per cent scenario actually demands

> **Every strategy requires effort well above the historical record — and even then, wages grow more slowly than the economy.**

STRATEGY	WHAT THE PROGRESSIVE SCENARIO REQUIRES	HOW THAT COMPARES WITH THE RECORD
Infrastructure	Infrastructure investment growing **25% a year** from 2025 to 2034	The 2014–2024 average, government and private combined, was about **10%**. This is more than twice as fast.
Digitalisation	Labour productivity gains of **1.3% a year** in metals rising to **3.4%** in machinery	Labour productivity already grows 3–6% a year. Technology adoption must make it **50% faster** than that.
Downstreaming	Upstream mineral exports diverted **8–12% a year**, downstream investment **+25% a year**, technology spillover **+1% a year**	Each 1% rise in capital raises downstream productivity by only 0.12%. **Without technology transfer the strategy does not pay.**
Agriculture	Productivity of leading regional crops **+3% a year**, processing capacity **+20% a year**	Requires productivity growth **50–100% faster** than the current 3–6% a year.
Green and blue	Freight 25% cheaper, household fuel spending shifted to public transport, energy efficiency **+2% a year**, maritime investment **+10% a year**	Cuts emissions **5.7% below baseline** and emissions intensity by 7.1%, without slowing growth.
Inclusiveness	Real wages grow more slowly than GDP in the baseline and in the conservative package	The progressive package **narrows the gap but does not close it**. Faster growth is more inclusive here, but it is not yet wage-led.

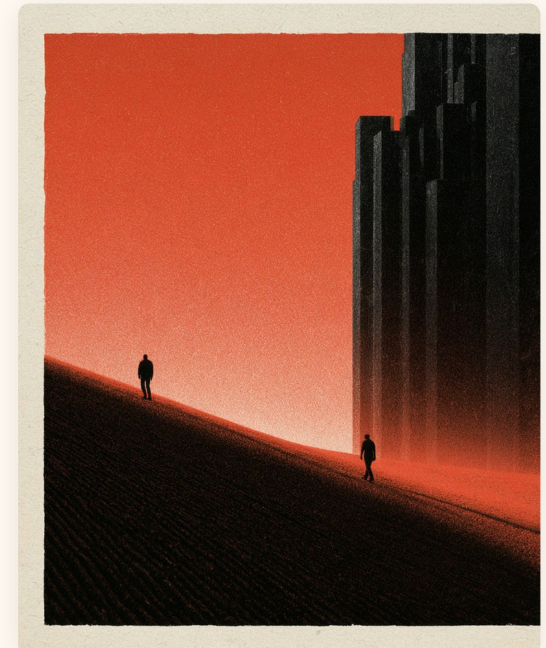
TWO REGIONS

Regional differences are unused potential, not an obstacle

Two provinces at opposite ends of Indonesia's structural transformation. Each was modelled separately, with the same model family.

> **A region early in its transformation can grow by catching up. A region at the frontier must find new engines. Neither route is the national average.**

	ACEH	KEPULAUAN RIAU
Income position	**Lower** middle income, while Indonesia as a whole is upper middle income	Among the **highest** incomes per head in Indonesia
Structure	Agrarian. Resembles Indonesia in the 1980s. Manufacturing share has fallen significantly.	Manufacturing, trade and logistics, built on the Batam–Bintan–Karimun industrial ecosystem.
Growth record	About **4%** a year, almost always below the national rate	Moderate, and dependent on established sectors
Position in the transformation	**Early.** Structural transformation has been delayed; agriculture is larger than its income level implies.	**At the frontier.** Exposed to global deindustrialisation and competition from China.
The question each faces	Can it accelerate without the export-manufacturing route?	Can it diversify without losing the engine it already has?



ACEH · THE AGRARIAN REGION

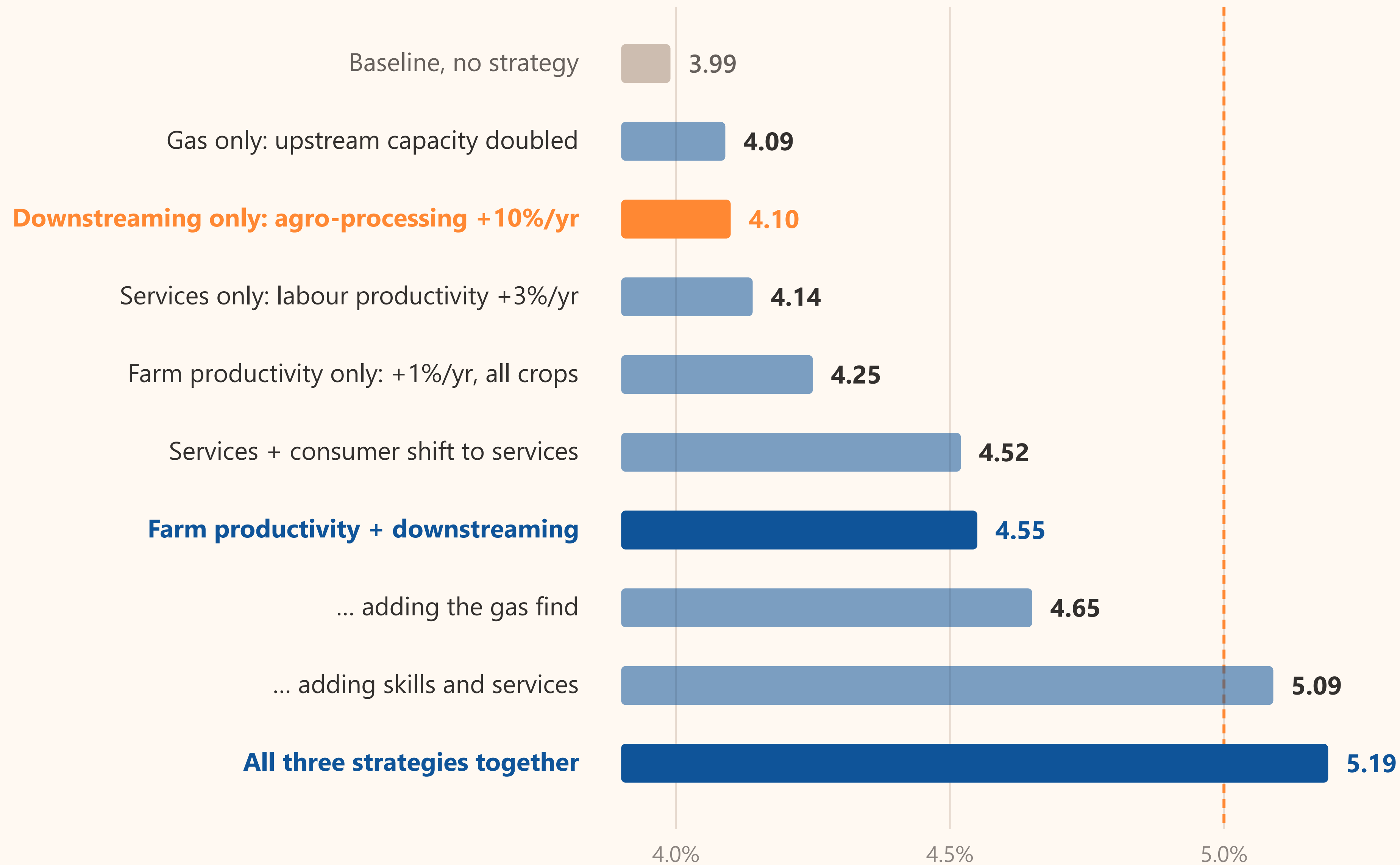
Downstreaming works, but not on its own

Average annual growth of Aceh's economy to 2040 under each strategy separately, then in combination.

> **Processing capacity alone adds 0.11 points. Combined with farm productivity it adds 0.56. All three strategies together lift Aceh above the national growth rate.**

Average annual growth of Aceh's economy, 2022–2040

IndoTERM Aceh · dashed line: the projected national growth rate, about 5%

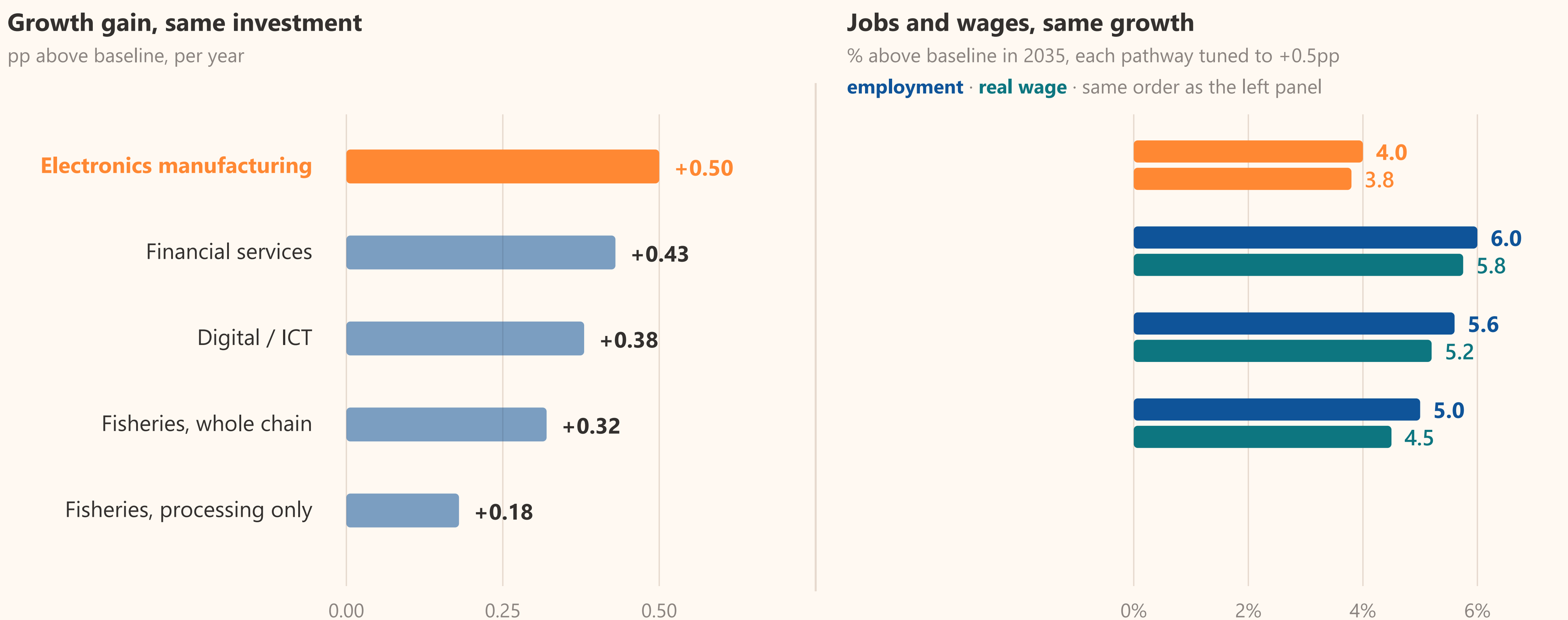


KEPULAUAN RIAU · THE INDUSTRIAL REGION

The strongest growth engine creates the fewest jobs

Left: what each pathway adds to growth for the same investment. Right: what that growth delivers in jobs and wages once every pathway is tuned to the same +0.5 points.

> **Electronics remains the most efficient way to buy growth. It is also the least inclusive: the alternative sectors create more jobs and higher real wages.**



SYNTHESIS

Six findings the three studies agree on

> **Three studies, one model family, six consistent findings.**

1

Capacity and productivity, together

No single strategy reaches the target. 8% required all five running at maximum simultaneously; in Aceh, no single scenario closed even half the gap. **Growth targets are met by packages, not by one flagship project.**

EGSA · ACEH

2

Downstreaming is real but modest on its own

Nationally it added 0.12–0.34 points, the second smallest of five levers. In Aceh, processing capacity alone added 0.11 — but **0.56 when paired with upstream productivity**. The pairing, not the plant, produces the growth.

EGSA · ACEH

3

Technology transfer is the condition

Downstreaming pays only with a productivity spillover of about 1% a year. Each 1% of new capital raises downstream productivity by 0.12%. **Investment without technology transfer buys capacity, not growth.**

EGSA

4

The right strategy depends on where the region stands

Aceh's agrarian structure is a disadvantage under an industry-led strategy and **an advantage under a multi-pronged one**. Kepri, at the frontier, needs the opposite move: diversification away from its strongest sector.

ACEH · KEPRI

5

Growth does not deliver inclusion by itself

Real wages grow more slowly than GDP in every national scenario. In Kepri every investment scenario lowered the inclusiveness index against baseline, and the fastest engine was the least inclusive. **Inclusion has to be designed in.**

EGSA · KEPRI

6

The region can be modelled before it is funded

All three studies used the same tool to compare strategies before money was committed. **A regional CGE model turns a strategy debate into a testable question** — including for a new economic area.

EGSA · ACEH · KEPRI

FOR TRANSMIGRATION AREAS

What this means for building an integrated economic area

Transmigration areas begin from the same position as Aceh: land-abundant, agrarian, early in the structural transformation, and below the national growth rate.

> **The evidence supports the direction of the transformation. It also says the sequence and the measurement matter more than the headline sector.**

1

Start from comparative advantage, measured

Identify each area's leading commodity and service sectors analytically — by linkage, transport cost, processing threshold and labour intensity — before committing investment. **The answer will differ by area.**

TOR: ANALYTICAL APPROACH

2

Build processing and farm productivity together

A processing plant without upstream productivity gains produced the weakest result in Aceh. **Raise yields and build capacity in the same programme**, not in sequence.

TRANS KARYA NUSA

3

Make technology transfer a condition of investment

The Morowali metal-processing polytechnic is the working model: investment arriving together with the training institution that transfers the technology. **Write the spillover into the deal.**

TRANS PATRIOT

4

Services are not the residual

In Aceh, skills and modern services added as much as agriculture and processing combined. In Kepri, services created the most jobs and the highest wages. **Treat the service sector as a growth engine, not overhead.**

TOR: LEADING SERVICE SECTORS

5

Measure jobs and real wages, not output

Output and inclusion diverged in every study. Publish indicators for employment, real wages and household consumption in the area alongside investment realised. **An area can succeed on output and still fail its residents.**

TOR: MEASURABLE INDICATORS

6

Build the evidence partnership now

Government, universities, industry and communities can maintain a standing regional model to test each strategy before it is funded, and to identify what is still unknown — there is **no CGE study of a transmigration area yet**.

TOR: RESEARCH GAPS, COLLABORATION

Sources. Bappenas and USAID Economic Growth Support Activity (EGSA), five white papers on infrastructure, digitalisation, mineral downstreaming, agricultural transformation and the green and blue economy, 2024, and their synthesis *Pendekatan Pemodelan untuk Strategi Pertumbuhan*. *Kajian Alternatif Sumber Pertumbuhan Ekonomi Aceh* (IndoTERM Aceh). *Pengembangan Model Computable General Equilibrium dan Kajian Alternatif Sumber Pertumbuhan Ekonomi Provinsi Kepulauan Riau* (IndoTERM Kepri). All simulations use the IndoTERM inter-regional CGE model, developed by CEDS Universitas Padjadjaran with the Centre of Policy Studies Victoria University, the Asian Development Bank and Bappenas.